Example 9.4: Let $f(x) = \tan x$. Then $f((-\frac{\pi}{2}, \frac{\pi}{2})) = (-\infty, +\infty)$.

Example 9.5: Let $f(x) = \cos x$. Then $f((-\infty, +\infty)) = [-1, 1]$.

Example 9.6: Let $f(x) = \arctan x$. Then $f((-\infty, +\infty)) = (-\frac{\pi}{2}, \frac{\pi}{2})$.

Example 9.7: Let $f(x) = e^x$. Then $f([0, +\infty)) = [1, +\infty)$.

The following theorem tells us that the image of a closed interval under a continuous function is always a closed interval:

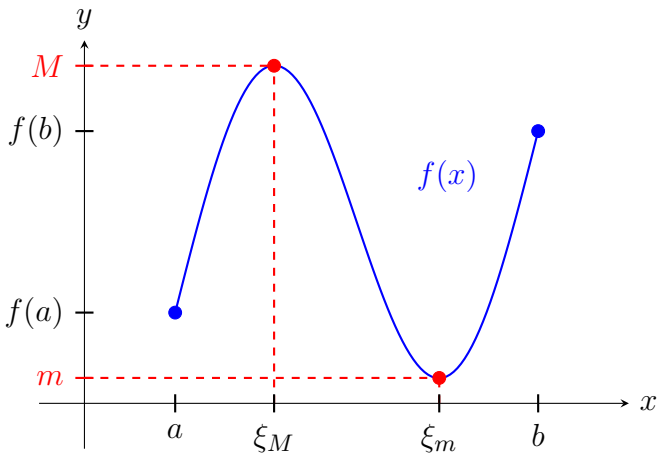
Weierstrass' Theorem

Let f be continuous on the interval $[a, b]$, where $a < b$ are real numbers. Then f is bounded on $[a, b]$ and it attains its minimum and maximum on $[a, b]$:

$$m = \min_{x\in[a,b]} f(x) \qquad \text{and} \qquad M = \max_{x\in[a,b]} f(x).$$

It follows that

$$f([a, b]) = [m, M].$$



Proof. **Step A.** The supremum. Define

$$M = \sup_{x\in[a,b]} f(x)$$

which can be a finite number or $+\infty$.

Case 1: M is finite. In this case, we know by definition of the supremum that for any $\varepsilon > 0$ there exists $x_\varepsilon \in [a, b]$ satisfying $M - \varepsilon < f(x_\varepsilon) \le M$. Our goal is to construct a sequence $\{x_n\}_{n\in\mathbb{N}_+}$, as follows: instead of ε take, for each $n \ge 1$, $\frac{1}{n}$, so that there exists $x_n \in [a, b]$ such that

$$M - \frac{1}{n} < f(x_n) \le M.$$

By the Squeeze Theorem for sequences (Theorem 6.15(5a)) it follows that

$$\lim_{n\to\infty} f(x_n) = M.$$

Case 2: M **is infinite.** In this case, again by the definition of the supremum, for each $n \in \mathbb{N}_+$ there exists $x_n \in [a, b]$ such that

$$f(x_n) > n.$$

By the Squeeze to $\pm\infty$ Theorem for sequences (Theorem 6.15(5b)) applied to $a_n = n \leq f(x_n) = b_n$ we must have

$$\lim_{n\to\infty} f(x_n) = +\infty = M.$$

Step B. The sequence $\{x_n\}_{n\in\mathbb{N}_+}$. In both cases, we obtained a sequence $\{x_n\}_{n\in\mathbb{N}_+} \subset [a, b]$. This is a bounded sequence (it is contained in a bounded interval). By the Bolzano-Weierstrass Theorem (Chapter 5) it has a convergent subsequence $\{x_{n_k}\}_{k\in\mathbb{N}_+}$. Call its limit ξ_M; since $x_n \in [a, b]$ for all n, we know (Proposition 5.3) that $\xi_M \in [a, b]$. Then we have:

$$\lim_{k\to\infty} x_{n_k} = \xi_M.$$

Moreover, since $\lim_{n\to\infty} f(x_n) = M$ (finite or $+\infty$), its subsequence $f(x_{n_k})$ has the same limit (see Section 5.2):

$$M = \lim_{n\to\infty} f(x_n) = \lim_{k\to\infty} f(x_{n_k}).$$

Using the continuity of f at the point ξ_M, we have

$$M = \lim_{k\to\infty} f(x_{n_k}) = f(\lim_{k\to\infty} x_{n_k}) = f(\xi_M).$$

But if $M = f(\xi_M)$ it must be a real number, so M cannot be $+\infty$. Furthermore, since M is attained at ξ_M, it belongs to the range of f on $[a, b]$, hence

$$M = \max_{x\in[a,b]} f(x).$$

The proof for m follows the same ideas.

Finally, the fact that $f([a, b]) = [m, M]$ is an immediate consequence of Corollary 9.5. □

Remark: The Bolzano-Weierstrass Theorem and Proposition 5.3 (a convergent sequence confined to a closed interval $[a, b]$ has its limit in $[a, b]$) were both established in Chapter 5; we use them repeatedly in this chapter.

> **Karl Weierstrass (1815–1897)**
>
> *Weierstrass is often called the father of modern analysis for the rigor he brought to the subject – the same ε-based rigor used throughout this course. He was also an unusually generous teacher. In the 1870s, Berlin's university authorities refused to let Sofia Kovalevskaya attend lectures, since women were barred from enrolling; rather than turn her away, Weierstrass tutored her privately, for free, for four years. He later called her his most talented pupil and pushed, successfully, for her to receive a doctorate – helping her become the first woman in Europe to hold one in mathematics.*

9.3 Invertibility of continuous functions

Lemma 9.6: Let f be continuous and invertible on an interval I. Let $x_1 < x_2 < x_3$ be points in I. Then exactly one of the following holds:

$$(i) \quad f(x_1) < f(x_2) < f(x_3)$$

or

$$(ii) \quad f(x_1) > f(x_2) > f(x_3).$$

Proof. Since f is invertible, it is 1-1. Therefore, since $x_1 \neq x_3$, also $f(x_1) \neq f(x_3)$.

Consider the case $f(x_1) < f(x_3)$ and assume *by contradiction* that (i) isn't satisfied. So $f(x_2) \notin (f(x_1), f(x_3))$.

Suppose that $f(x_2)$ lies to the right of the interval $(f(x_1), f(x_3))$, so that $f(x_1) < f(x_3) < f(x_2)$. Consider f on $[x_1, x_2]$. Since it is continuous there, by Theorem 9.4 (The Intermediate Value Theorem), $f|_{[x_1,x_2]}$ must assume all values in $[f(x_1), f(x_2)]$. Since $f(x_1) < f(x_3) < f(x_2)$, there will be $z \in [x_1, x_2]$ such that $f(z) = f(x_3)$. But this contradicts the fact that f is 1-1. Hence (i) is satisfied and $f(x_1) < f(x_2) < f(x_3)$.

The case that $f(x_2)$ lies to the left of the interval $(f(x_1), f(x_3))$ is handled in a similar way.

The proof for the case $f(x_1) > f(x_3)$ follows the same ideas, and the lemma follows. □

Theorem 9.7: Let f be continuous on an interval $I \subseteq \mathbb{R}$. Then

$$f \text{ is 1-1 on } I \quad \Leftrightarrow \quad f \text{ is strictly monotone on } I.$$

Proof. The direction $\Leftarrow$ was already proven in Proposition 4.1.

Hence we only need to prove the implication $\Rightarrow$. Recall that a 1-1 function is invertible, so that it is enough to prove

$$f \text{ is invertible on } I \quad \Rightarrow \quad f \text{ is strictly monotone on } I.$$

Let $x_1 < x_2$ be points in I. We'll show that if $f(x_1) < f(x_2)$ then f is strictly increasing on $[x_1, x_2]$. Let z_1, z_2 be two points such that

$$x_1 \leq z_1 < z_2 \leq x_2.$$

We want to show that $f(z_1) < f(z_2)$. If $z_1 = x_1$ or $z_2 = x_2$, this follows from Lemma 9.6 applied to the trio $x_1 < z_2 < x_2$ or $x_1 < z_1 < x_2$ (or directly, if both hold). So assume $x_1 < z_1 < z_2 < x_2$.

Using Lemma 9.6 for the trio $x_1 < z_1 < x_2$, we find that

$$f(x_1) < f(z_1) < f(x_2).$$

Using Lemma 9.6 for the trio $z_1 < z_2 < x_2$ (alternative (ii) is impossible since $f(z_1) < f(x_2)$), we find that

$$f(z_1) < f(z_2) < f(x_2).$$